Final Project EDT

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May 2026

1 Phase 1: Problem Definition and Data Acquisition

1.1 Event Definition and Study Area

This project revisits the case study analyzed in the Midterm Project: the collapse of the Kakhovka Dam on June 6, 2023, located on the Dnieper River in Kherson Oblast, southern Ukraine. The event caused catastrophic downstream flooding and the rapid drainage of the Kakhovka Reservoir, making it one of the most significant hydrological disasters in recent European history.

The Region of Interest (ROI) is centered on the area surrounding Nova Kakhovka, where the effects of reservoir drainage were most clearly observable in satellite imagery. This ROI was selected based on image availability and the visibility of hydrological changes in the post-event date (June 18, 2023).

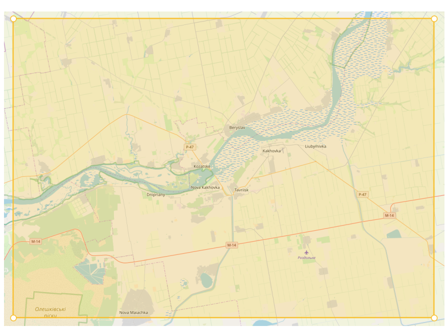

Figure 1: ROI on an OpenStreetMap background.

Figure 2: ROI on Sentinel-2 image post-event.

1.2 Satellite Imagery Metadata

Two Sentinel-2 Level-2A images were acquired from the Copernicus Open Access Hub. Table 1 summarizes their main technical characteristics.

Table 1: Technical characteristics of the satellite imagery used.

Date	Satellite	Level	Resolution	Cloud Cover	Role
June 18, 2023	Sentinel-2 A/B	Level-2A	10 m	<30%	Post-event

Level-2A products provide Bottom-of-Atmosphere (BOA) reflectance, eliminating the need for additional atmospheric correction. The bands used for this project span from the visible to the shortwave infrared spectrum, as detailed in Table 2.

Table 2: Sentinel-2 spectral bands used for dataset construction.

Band	Name	Wavelength (μm)
B2	Blue	0.490
B3	Green	0.560
B4	Red	0.665
B5	Red Edge 1	0.705
B6	Red Edge 2	0.740
B7	Red Edge 3	0.783
B8	Near-Infrared (NIR)	0.842
B8A	Narrow NIR	0.865
B11	SWIR 1	1.610
B12	SWIR 2	2.190

1.3 Dataset Construction

To prepare the training dataset for the machine learning phase, pixel values were extracted from the Sentinel-2 band stack at the locations defined by the training polygons digitized in QGIS. Each polygon corresponds to one of four land cover classes: Water, Vegetation, Bare Soil, and Built-up.

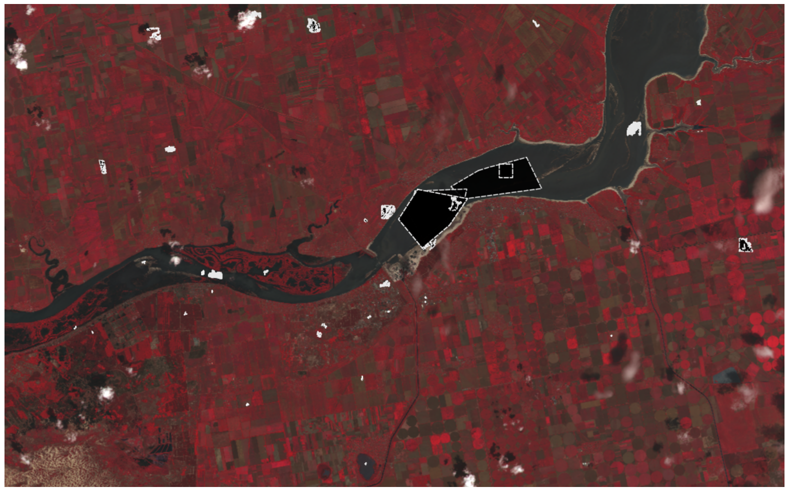

Figure 3: False-color composite (NIR–Red–Green) with digitized training polygons over the post-event study area.

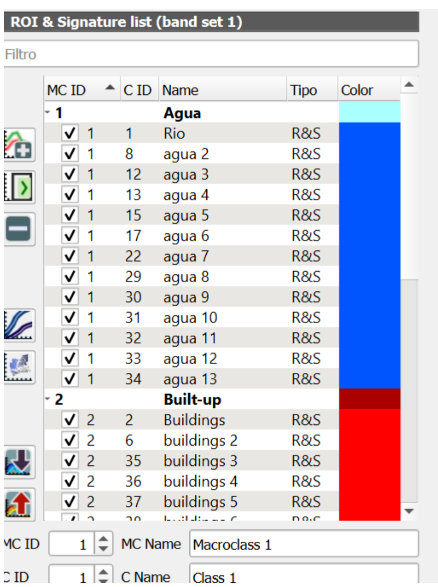

Figure 4: ROI & Signature list: clases Water (MC 1) y Built-up (MC 2).

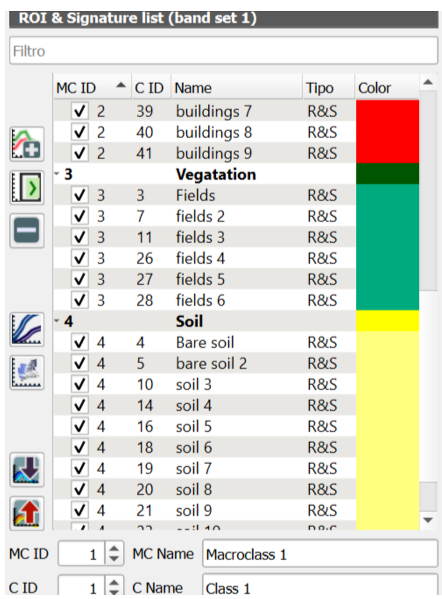

Figure 5: ROI & Signature list: clases Vegetation (MC 3) y Soil (MC 4).

The extraction was performed using a Python script based on the `rasterio` and `geopandas` libraries. For each pixel falling within a training polygon, the script records its geographic coordinates (Latitude, Longitude), the reflectance values for all ten spectral bands, and the corresponding class label. The output was exported as a Tab-Separated Values (TSV) file, as required for the subsequent training phase.

Table 3 illustrates the structure of the resulting dataset.

Table 3: Structure of the exported TSV training dataset.

Column	**Type**	**Description**
Latitude	Float	Geographic latitude of the pixel centroid
Longitude	Float	Geographic longitude of the pixel centroid
B2	Float	Blue band reflectance
B3	Float	Green band reflectance
B4	Float	Red band reflectance
B5	Float	Red Edge 1 reflectance
B6	Float	Red Edge 2 reflectance
B7	Float	Red Edge 3 reflectance
B8	Float	NIR band reflectance
B8A	Float	Narrow NIR reflectance
B11	Float	SWIR 1 reflectance
B12	Float	SWIR 2 reflectance
label	String	Land cover class (Water, Vegetation, Bare Soil, Built-up)

Table 4 shows the first three extracted pixels for each land cover class.

Table 4: Sample of the training dataset: first three pixels per class (band values in DN).

Latitude	**Longitude**	**B2**	**B3**	**B4**	**B5**	**B6**	**B7**	**B8**	**B8A**	**B11**	**B12**	**Label**
46.7608	33.3754	2194	2594	2814	3081	3493	3680	3706	3814	3981	3641	Built-up
46.7613	33.3728	2228	2422	2636	3922	4128	4250	3146	4389	5874	6179	Built-up
46.7432	33.3801	3118	3508	3654	3765	3625	3636	3614	3530	4830	5244	Built-up
46.7859	33.7286	1672	2032	1980	2582	3697	3947	4010	4147	4130	3402	Soil
46.9410	33.3038	1637	1803	2108	2274	2362	2492	2582	2731	4313	4164	Soil
46.9398	33.3070	1618	1822	2116	2325	2395	2520	2564	2732	4246	4110	Soil
46.8116	33.3766	1433	1647	1747	2160	3027	3297	3254	3573	3201	2323	Vegetation
46.8139	33.3806	1418	1698	1732	2232	3184	3495	3654	3789	3365	2495	Vegetation
46.8154	33.3775	1418	1693	1552	2172	3293	3501	3520	3749	2974	2121	Vegetation
46.8072	33.4318	1930	2158	2342	2509	2602	2727	2814	2894	2539	1949	Water
46.8025	33.4031	1750	1896	1952	2074	2017	2046	2130	1998	1588	1474	Water
46.7926	33.4182	1693	1872	2082	2246	2365	2422	2532	2712	4124	2849	Water

2 Phase 2: Pre-processing and Training

2.1 Pre-processing

Before training the machine learning models, several pre-processing operations were applied to the dataset in order to improve data consistency, ensure fair model evaluation, and optimize classification performance.

The exported TSV dataset contained spectral reflectance values extracted from Sentinel-2 imagery for four land cover classes: Built-up, Soil, Vegetation, and Water. Each sample represented a single pixel and included ten spectral features corresponding to bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12.

The initial extraction yielded a highly imbalanced dataset of 547,173 pixels, with a severe class disparity as shown in Table 5.

Table 5: Raw pixel distribution before balancing.

Class	**Pixel count**
Water	510,071
Soil	22,171
Vegetation	13,379
Built-up	1,552
Total	547,173

Water dominated the dataset with over 93% of all pixels, which would have caused severe bias toward that class in any trained model. To correct this, additional training polygons were digitized specifically for the Built-up class, and the dataset was subsequently balanced by random undersampling to 2,027 pixels per class, yielding a final balanced dataset of 8,108 samples evenly distributed among the four categories.

Prior to model training, the dataset was inspected for missing or invalid values. Since the spectral extraction process was performed directly from Sentinel-2 Level-2A products, no significant missing data were detected and no additional interpolation procedures were required.

The categorical class labels were encoded into numerical representations in order to be compatible with the machine learning algorithms. Subsequently, the dataset was divided into training and testing subsets using stratified sampling to preserve the original class distribution. Approximately 70% of the samples were used for training and 30% for testing.

Feature scaling was also applied during the pre-processing stage. Since some algorithms, particularly Artificial Neural Networks (ANN) and K-Nearest Neighbors (KNN), are sensitive to differences in feature magnitude, the spectral bands were normalized using standard scaling. This transformation ensured that all features contributed proportionally during the optimization process.

For the ANN implementation, the target labels were additionally transformed into categorical representations suitable for multiclass classification with a softmax output layer. Regularization techniques such as Dropout were later used during training to reduce the risk of overfitting.

Overall, the pre-processing stage ensured that the spectral dataset was properly formatted, balanced, normalized, and ready for supervised land cover classification using the selected machine learning approaches.

2.2 Models

2.2.1 Decision Trees (DT)

The following table summarizes the hyperparameter configurations tested during the model optimization process.

Table 6: Hyperparameter search space for the Decision Tree classifier.

Hyperparameter	**Values tested**
`criterion`	gini, entropy, log_loss
`max_depth`	None, 5, 10, 15, 20
`min_samples_split`	2, 5, 10
`min_samples_leaf`	1, 2, 4
Method: 5-fold StratifiedKFold, scoring: F1-macro	

The results show excellent performance of the Decision Tree model in the post-event classification problem. An overall accuracy of 98.56% means that, out of 2433 test samples, almost all were correctly classified. Furthermore, a Kappa coefficient of 0.9808 indicates very high agreement beyond chance, confirming that the model is actually learning useful spectral patterns rather than simply exploiting the class distribution.

Table 7: Classification Report for the Decision Tree Model

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	0.9806	0.9951	0.9878	608
Soil	0.9867	0.9737	0.9801	608
Vegetation	0.9870	0.9984	0.9927	609
Water	0.9883	0.9753	0.9818	608

The macro F1-score of 0.9856 reinforces this conclusion because it averages performance across classes while giving equal importance to each one, and it still remains very high. Likewise, the weighted F1-score of 0.9856 closely matches the macro value, suggesting balanced performance and the absence of clearly disadvantaged classes. In other words, the model not only performs well globally, but also consistently across all land-cover categories.

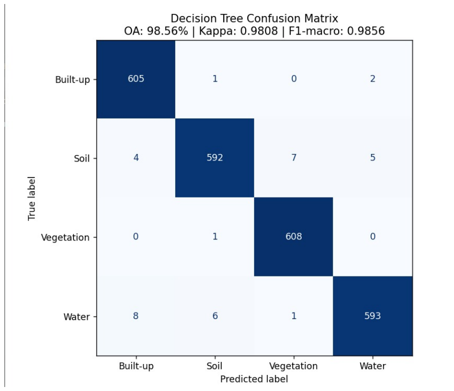

Figure 6: Confusion matrix obtained from the Decision Trees.

The confusion matrix shows that *Vegetation* is the strongest class (608/609 correctly classified samples), followed by *Built-up* (605/608). The most frequent confusions occur between *Soil* and *Water*, and to a lesser extent between *Water* and *Built-up*. This behavior is expected in satellite imagery because of local spectral similarities, mixed pixels, and humidity or shadow effects. Even with these specific confusions, the number of errors is very small compared to the total number of samples and does not compromise the overall quality of the model.

Finally, it can be concluded that the Decision Tree model provides a good performance for the Kakhovka scene and achieves effective land-cover separation.

2.2.2 Support Vector Machine (SVM)

A Support Vector Machine (SVM) classifier was implemented to perform supervised land cover classification using the spectral information extracted from the Sentinel-2 imagery. SVM is a widely used machine learning algorithm in remote sensing applications due to its strong performance in high-dimensional datasets and its ability to separate classes using optimal decision boundaries.

The training dataset, evenly distributed among the four land cover classes: Built-up, Soil, Vegetation, and Water. Each sample corresponds to a single pixel and contains the reflectance values from the ten spectral bands used in this project.

Before training, the dataset was divided into training and testing subsets using a stratified split in order to preserve the class distribution. Approximately 70% of the samples were used for training and 30% for evaluation.

Table 8: Hyperparameter search space for the SVM classifier.

Hyperparameter	**Values tested**
`kernel`	RBF (fixed)
`C`	0.5, 1.0, 1.5, ..., 15.0 (step 0.5, 30 values)
`gamma`	scale (fixed)
Method: sequential evaluation on test set	

The SVM model achieved an Overall Accuracy of 99.75%, demonstrating excellent classification performance across all land cover categories. Precision, recall, and F1-score values were close to 1.00 for every class, indicating that the spectral signatures extracted from the Sentinel-2 bands provide strong class separability.

Table 9 summarizes the evaluation metrics obtained during testing.

Table 9: Classification report for the SVM model.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	1.00	1.00	1.00	608
Soil	1.00	0.99	1.00	608
Vegetation	0.99	1.00	1.00	609
Water	1.00	1.00	1.00	608

The confusion matrix shown in Figure 7 confirms the high classification accuracy of the model. Most samples were correctly classified, with only a very small number of misclassifications between spectrally similar classes.

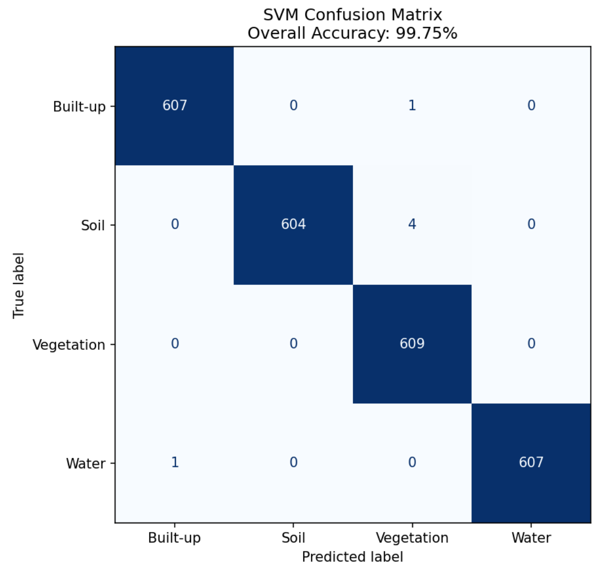

Figure 7: Confusion matrix obtained from the SVM classifier.

These results indicate that the SVM model is highly effective for multispectral land cover classification using Sentinel-2 imagery and provides a strong baseline for comparison with the other machine learning approaches evaluated in this project.

2.2.3 Artificial Neural Networks (ANN)

An Artificial Neural Network (ANN) model was implemented to evaluate the performance of deep learning techniques for multispectral land cover classification. Neural networks are capable of learning complex nonlinear relationships between spectral features and target classes, making them highly suitable for remote sensing applications involving high-dimensional data.

The ANN was trained using the same dataset employed for the SVM classifier, containing 8108 labeled pixel samples equally distributed among the classes Built-up, Soil, Vegetation, and Water. Each sample contains the reflectance values corresponding to the ten Sentinel-2 spectral bands selected for this study.

Prior to training, the dataset was normalized and divided into training and testing subsets using a stratified split. Approximately 70% of the data was used for training, while the remaining 30% was reserved for model evaluation.

The neural network architecture consisted of multiple fully connected dense layers with ReLU activation functions and Dropout regularization layers to reduce overfitting. The final output layer used a softmax activation function to perform multiclass classification across the four land cover categories.

Table 10: Hyperparameter search space for the ANN classifier.

Hyperparameter	**Values tested**
`learning_rate`	0.1, 0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001
`optimizer`	Adam (fixed)
`epochs`	50 (fixed)
`batch_size`	32 (fixed)
`dropout`	0.3 (fixed)
`architecture`	Dense(64)–Dropout–Dense(128)–Dropout–Dense(64)–Softmax
Method: sequential evaluation on test set	

The model was trained for 50 epochs and achieved a final Overall Accuracy of 99.59% on the testing dataset. The training history demonstrated stable convergence and high validation accuracy throughout the training process.

Table 11 summarizes the classification metrics obtained for the ANN model.

Table 11: Classification report for the ANN model.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	1.00	1.00	1.00	608
Soil	0.99	1.00	0.99	608
Vegetation	1.00	1.00	1.00	609
Water	1.00	0.99	1.00	608

Figure 8 presents the confusion matrix generated from the ANN predictions. The matrix indicates that the vast majority of samples were correctly classified, with only minimal confusion between a few spectrally similar classes.

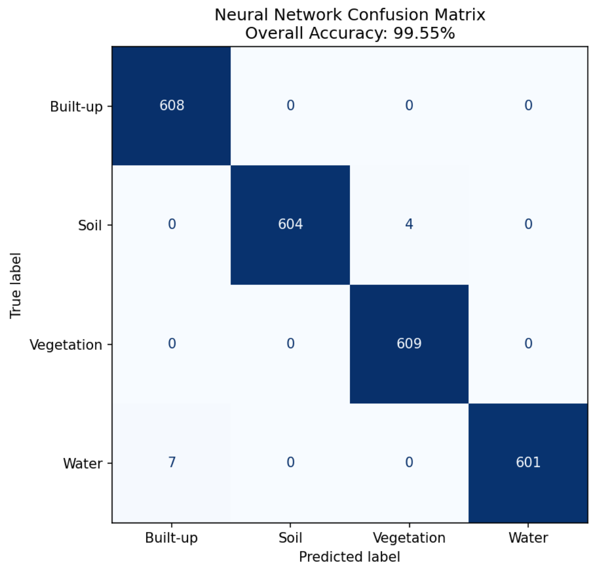

Figure 8: Confusion matrix obtained from the ANN classifier.

Figure 9 shows the training history of the neural network, including the evolution of training and validation accuracy during the learning process. The results demonstrate rapid convergence and stable generalization performance.

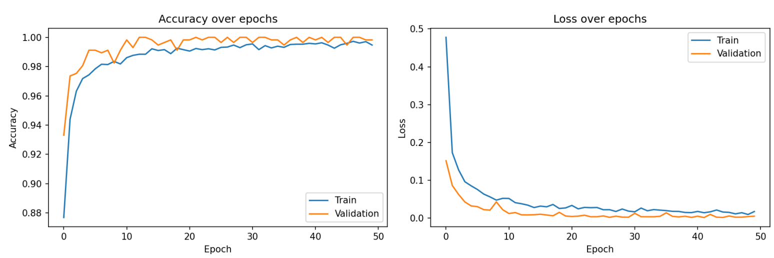

Figure 9: Training and validation accuracy history for the ANN model.

The ANN model achieved performance comparable to the SVM classifier, confirming that deep learning approaches can effectively model the spectral behavior of different land cover types using Sentinel-2 multispectral imagery.

2.2.4 K-Nearest Neighbor (KNN)

The KNN classifier was trained using an 80/20 train-test split with stratification across all four land cover classes. Prior to training, all spectral band values were normalized using standard scaling, as KNN relies on Euclidean distance and is sensitive to differences in feature magnitude. Hyperparameter optimization was performed via 5-fold cross-validation over a grid of candidate values for the number of neighbors, weighting scheme, and distance metric.

Table 12: Hyperparameter search space for the KNN classifier.

Hyperparameter	**Values tested**
`n_neighbors`	3, 5, 7, 9, 11
`weights`	uniform, distance
`metric`	Euclidean (fixed)
Method: 5-fold GridSearchCV, scoring: F1-macro	

The optimal configuration found was $K = 3$, uniform weighting, and Euclidean distance, achieving a macro-averaged F1-Score of 0.9943 during cross-validation. Table 13 summarizes the per-class performance metrics obtained on the test set, and Table 14 presents the corresponding confusion matrix.

Table 13: Per-class performance metrics for the KNN classifier.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
Built-up	0.9975	0.9951	0.9963	406
Soil	0.9975	0.9926	0.9950	405
Vegetation	0.9927	1.0000	0.9963	406
Water	0.9975	0.9975	0.9975	405
Overall Accuracy	99.63%			
Kappa Coefficient	0.9951			

Table 14: Confusion matrix for the KNN classifier (rows = predicted, columns = reference).

	Built-up	**Soil**	**Vegetation**	**Water**
Built-up	404	1	1	0
Soil	0	402	2	1
Vegetation	0	0	406	0
Water	1	0	0	404

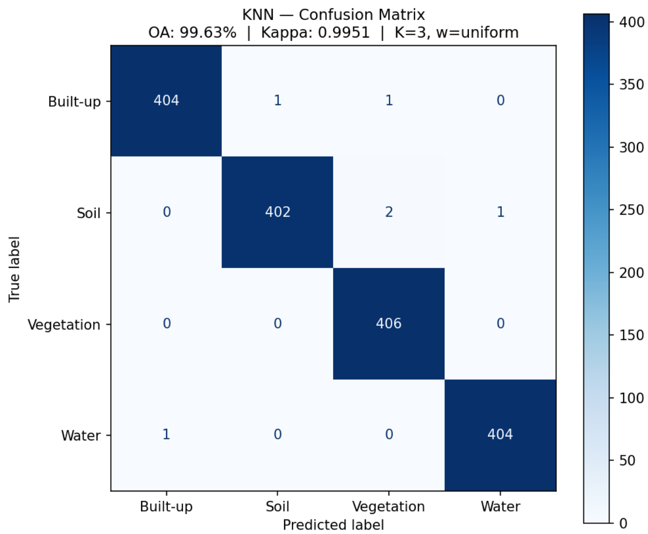

Figure 10: Confusion matrix for the KNN classifier ($K = 3$, uniform weights, Euclidean distance). OA = 99.63%, Kappa = 0.9951.

2.2.5 Naive Bayes (NB)

The fifth supervised learning architecture implemented in this project was the Naive Bayes classifier. This model is based on Bayes' theorem and estimates the probability that a pixel belongs to a given land-cover class according to its spectral response. In this case, the input variables correspond to the Sentinel-2 spectral bands extracted during the dataset construction stage.

Since the spectral bands are continuous numerical variables, the selected implementation was Gaussian Naive Bayes. This variant assumes that the values of each feature follow a Gaussian distribution within each class. Although the model assumes conditional independence between the spectral bands, which is a strong simplification for multispectral satellite data, it provides a useful probabilistic baseline for comparison against more complex models such as Decision Trees, Support Vector Machines, Artificial Neural Networks, and K-Nearest Neighbors.

The model was trained using the tabular dataset generated from the training polygons. The input features were the spectral bands B2, B3, B4, B5, B6, B7, B8, B8A, B11, and B12, while the target variable was the land-cover class label. The dataset was divided into training and testing subsets, preserving the class distribution. The final split contained 5675 training samples and 2433 testing samples.

A hyperparameter search was performed over the `var_smoothing` parameter. The best value obtained was:

$$\texttt{var\_smoothing} = 1 \times 10^{-12} \tag{1}$$

This value was used to train the final Gaussian Naive Bayes model.

Table 15: Summary metrics for the Gaussian Naive Bayes classifier.

Metric	**Value**
Best `var_smoothing`	1.0×10^{-12}
Cross-validation F1 macro	0.8518
Overall accuracy	0.8442
Kappa coefficient	0.7923
F1 macro	0.8444
F1 weighted	0.8445
Training samples	5675
Testing samples	2433

The obtained overall accuracy was 0.8442, meaning that approximately 84.42% of the test pixels were correctly classified. The Kappa coefficient reached 0.7923, indicating a strong level of agreement between the predicted and reference classes beyond random chance. The F1 macro score was 0.8444, showing that the classifier achieved balanced performance across the four land-cover categories.

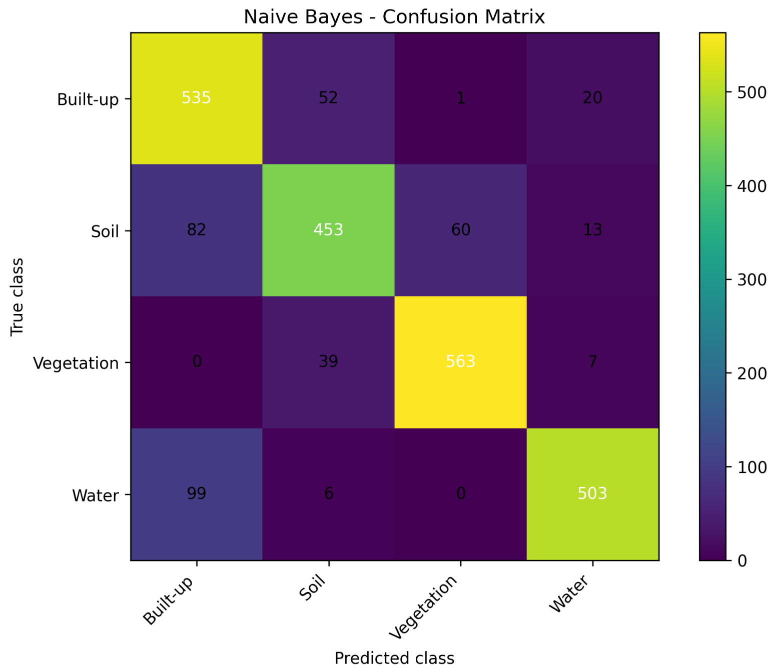

Figure 11: Confusion matrix obtained with the Gaussian Naive Bayes classifier.

Table 16: Confusion matrix for the Gaussian Naive Bayes classifier.

True / Predicted	**Built-up**	**Soil**	**Vegetation**	**Water**
Built-up	535	52	1	20
Soil	82	453	60	13
Vegetation	0	39	563	7
Water	99	6	0	503

The confusion matrix shows that the model performed well for all classes, with the highest number of correct predictions located along the main diagonal. Vegetation was the best classified class, with 563 correctly predicted pixels and an F1-score of 0.9132. This indicates that vegetation has a clearly distinguishable spectral signature in the selected Sentinel-2 bands.

Water also obtained strong results, with 503 correctly classified pixels and an F1-score of 0.8740. However, 99 water pixels were incorrectly classified as built-up areas. This confusion may be associated with spectral similarities in dark or low-reflectance regions, shadows, or mixed pixels near the boundaries between flooded and urbanized zones.

Built-up areas achieved 535 correct predictions and an F1-score of 0.8082. The main confusion for this class occurred with soil, where 52 built-up pixels were classified as soil. This behavior is expected in post-event satellite imagery, since exposed surfaces, debris, dry soil, and artificial materials may present similar spectral responses.

The soil class obtained the lowest F1-score among the four classes, with a value of 0.7824. The main errors occurred between soil and built-up areas, with 82 soil pixels classified as built-up, and between soil and vegetation, with 60 soil pixels classified as vegetation. This suggests that soil is the most spectrally ambiguous class in the current dataset.

Table 17: Classification report for the Gaussian Naive Bayes classifier.

Class	**Precision**	**Recall**	**F1-score**	**Support**
Built-up	0.7472	0.8799	0.8082	608
Soil	0.8236	0.7451	0.7824	608
Vegetation	0.9022	0.9245	0.9132	609
Water	0.9263	0.8273	0.8740	608
Macro avg.	0.8499	0.8442	0.8444	2433
Weighted avg.	0.8499	0.8442	0.8445	2433

Overall, the Gaussian Naive Bayes classifier provided a solid probabilistic baseline for land-cover classification in the Kakhovka post-event scene. Its main advantage is that it is computationally simple and fast to train, while still producing competitive results. Nevertheless, some confusion remains between spectrally similar classes, particularly soil and built-up areas, as well as water and built-up areas. These limitations are consistent with the independence assumption of Naive Bayes, since the model does not explicitly capture correlations between spectral bands.

Therefore, the Naive Bayes model can be considered a useful reference model for the comparative evaluation stage. Its final performance, with an overall accuracy of 0.8442, a Kappa coefficient of 0.7923, and a macro F1-score of 0.8444, provides a baseline against which the remaining supervised classifiers can be compared.

3 Phase 3: Synthesis and Prediction

3.1 Model Comparison

Table 18 summarizes the overall performance of all five supervised classifiers evaluated in this project. The metrics reported correspond to the test set in each case.

Table 18: Comparative performance of all classifiers on the test set.

Model	**OA (%)**	**Kappa**	**F1-macro**	**F1-weighted**	**Best hyperparameter**
Decision Tree	98.56	0.9808	0.9856	0.9856	entropy, depth=None
SVM	99.75	0.9967	0.9975	0.9975	C = 9.5
ANN	99.59	0.9945	0.9959	0.9959	lr = 0.005
KNN	99.63	0.9951	0.9963	0.9963	K=3, uniform
Naive Bayes	84.42	0.7923	0.8444	0.8445	var_smoothing=10^{-12}

Among the five models evaluated, SVM, ANN, and KNN achieved the highest overall accuracies (above 99%), demonstrating that the spectral signatures extracted from the ten Sentinel-2 bands provide strong class separability for this scene. The Decision Tree also performed competitively with an OA of 98.56%, while remaining fully interpretable. Naive Bayes was the weakest performer at 84.42%, consistent with its independence assumption between spectral bands, which is a known limitation for multispectral data.

The main confusion across all models occurred between spectrally similar classes, particularly Soil and Built-up, and to a lesser extent Water and Built-up in the post-event context. These errors are expected given the spectral ambiguity of exposed surfaces and debris in post-flood satellite imagery.